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## Conditioning & Transformations of a RV

EEE 554 — Lecture #6

### Conditioning a RV on an Event

Let  $A$  be an event defined in terms of a random variable (RV)  $x$ ; e.g.,  $A = \{0 < x < 1\}$  or  $A = \{|x| < 5\}$ . The conditional PDF of  $x$  given  $A$  is:

$$f_x(x|A) = \begin{cases} \frac{f_x(x)}{P(A)} & x \in A \\ 0 & x \notin A \end{cases}$$

Note that  $f_x(x|A) \geq 0$  for all  $x$  and

$$\int_{-\infty}^{\infty} f_x(x|A)dx = \frac{1}{P(A)} \int_A f_x(x)dx = 1,$$

so  $f_x(x|A)$  is a valid PDF.

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### Conditional Moments

Moments are defined in the usual way with respect to the conditional PDF:

$$E[x|A] = \int_{-\infty}^{\infty} x f_x(x|A)dx$$

And the expectation rule applies:

$$E[g(x)|A] = \int_{-\infty}^{\infty} g(x) f_x(x|A)dx$$

So,

$$E[x^n|A] = \int_{-\infty}^{\infty} x^n f_x(x|A)dx$$

and

$$\text{var}(x|A) = E[x^2|A] - (E[x|A])^2$$

**Example:**

$$f_x(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases}$$

Let  $A = \{x > \frac{1}{2}\} \implies P(A) = \frac{3}{4}$ .

The conditional PDF is:

$$f_x(x|A) = \begin{cases} \frac{8}{3}x & \frac{1}{2} < x \leq 1 \\ 0 & \text{else} \end{cases}$$

Calculating conditional moments:

$$E[x|A] = \frac{8}{3} \int_{1/2}^1 x^2 dx = \frac{8}{9} x^3 \Big|_{1/2}^1 = \frac{8}{9} \left(1 - \frac{1}{8}\right) = \frac{7}{9}$$

$$E[x^2|A] = \frac{8}{3} \int_{1/2}^1 x^3 dx = \frac{2}{3} x^4 \Big|_{1/2}^1 = \frac{2}{3} \left(1 - \frac{1}{16}\right) = \frac{5}{8}$$

$$\text{var}(x|A) = \frac{5}{8} - \left(\frac{7}{9}\right)^2 = \frac{5}{8} - \frac{49}{81} = \frac{13}{648}$$


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## Transformation of One RV

Suppose  $g(x)$  is a monotonically increasing function. Define a new RV  $y = g(x)$ . Then:

$$\begin{aligned} F_y(y) &= P(y \leq y) = P(g(x) \leq y) \\ &= P(x \leq g^{-1}(y)) = F_x(g^{-1}(y)) \end{aligned}$$

where  $g^{-1}$  is defined so that  $g(g^{-1}(y)) = y$ .

Note that by the chain rule:

$$\frac{d}{dy} g(g^{-1}(y)) = g'(g^{-1}(y)) \cdot (g^{-1})'(y) = 1$$

So,

$$(g^{-1})'(y) = \frac{1}{g'(g^{-1}(y))}$$

Thus, the PDF of  $y$  is:

$$\begin{aligned} f_y(y) &= \frac{d}{dy} F_y(y) = \frac{d}{dy} F_x(g^{-1}(y)) \\ &= f_x(g^{-1}(y)) \cdot (g^{-1})'(y) \\ &= \frac{f_x(x)}{g'(x)} \Big|_{x=g^{-1}(y)} \end{aligned}$$

**Example:**  $x \sim U[-1, 1]$  (Uniform), and  $y = g(x) = x^3$ . We have  $g^{-1}(y) = y^{1/3}$  and  $g'(x) = 3x^2$ .

$$f_y(y) = \frac{(1/2)}{3x^2} \Big|_{x=y^{1/3}} = \frac{1}{6} y^{-2/3}$$

for  $-1 < y < 1$  and zero otherwise.

Note that it integrates to 1:

$$\int_{-\infty}^{\infty} f_y(y) dy = \frac{1}{6} \int_{-1}^1 y^{-2/3} dy = \frac{1}{2} y^{1/3} \Big|_{-1}^1 = 1$$

With this PDF for  $y$ , direct calculation yields:

$$E[y^2] = \int_{-\infty}^{\infty} y^2 f_y(y) dy = \frac{1}{6} \int_{-1}^1 y^{4/3} dy = \frac{1}{7}$$

Using the expectation rule with  $f_x$  gives the same result:

$$E[y^2] = E[x^6] = \frac{1}{2} \int_{-1}^1 x^6 dx = \frac{1}{7}$$

If  $g$  is monotonically **decreasing**, a similar calculation to the one above gives:

$$f_y(y) = \frac{f_x(x)}{|g'(x)|} \Big|_{x=g^{-1}(y)}$$

If  $g$  is not strictly monotonic everywhere but is monotonic on a sequence of disjoint intervals  $A_k = [a_k, b_k)$ , then by total probability:

$$f_y(y) = \sum_k f_y(y|A_k)P(A_k)$$

where each  $f_x(x|A_k)$  is monotonic. The expression may be applied to each of these conditional PDFs individually, which can be combined to obtain  $f_y(y)$ .

## Linear Transformation of a Normal RV

Let  $x \sim N(\mu, \sigma^2)$  and set:

$$y = g(x) = ax + b$$

with  $a \neq 0$ . Then  $g^{-1}(y) = \frac{y-b}{a}$  and  $|g'(x)| = |a|$ .

Using the transformation rule:

$$\begin{aligned} f_y(y) &= \frac{f_x(x)}{|a|} \Big|_{x=\frac{y-b}{a}} \\ f_y(y) &= \frac{1}{|a|} \cdot \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} \left( \frac{y-b}{a} - \mu \right)^2 \right\} \\ &= \frac{1}{\sqrt{2\pi}(a^2\sigma^2)} \exp \left\{ -\frac{1}{2(a^2\sigma^2)} (y - (a\mu + b))^2 \right\} \end{aligned}$$

So,  $y \sim N(a\mu + b, a^2\sigma^2)$ ; i.e., a linear transformation on a normal RV produces another normal RV.

Even without calculating  $f_y$  explicitly, it is obvious that:

$$E[y] = a\mu + b \quad \text{and} \quad \text{var}(y) = a^2\sigma^2$$

Hence, knowing  $y$  is normal immediately gives its PDF.